



Sony SmartWatch 2 is here!

M Sony has launched its latest flagship smartphone Xperia Z1 and the SmartWatch 2 in India <http://dgit.in/1a39DSb>

Surface 2 announced

Microsofts claims to have worked on most of the drawbacks of its predecessor <http://dgit.in/18UkM4H>

The Bleeding Edge: Tech from the MIT Media Labs

The MIT Media Labs is a place where the brightest, young minds get together and work on some amazing projects that could very well shape our future. Here are some of the projects that blew our minds.

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Biological 3D printing with the help of silkworms

Researchers from the MIT Media Lab's Mediated Matter Group have built a large dome measuring 12 by 12 feet with the help of 6,500 live silkworms. The dome, branded 'Silk Pavilion', was constructed after researchers studied silkworms to understand the process behind how the creatures created silk. The researchers refer to the silkworm project as "a biological swarm approach to 3D printing" whose striking element is the absence of homogeneity in the construction process.

The benefit of non-homogenous 3D printing, according to the researchers, is that it is able to apply non-uniform proper-



The silk dome created with the help of 6,500 silkworms

ties to the object being created based on the environment in which it's being created and according to the varied needs of the creator. For e.g. a silkworm's cocoon will be rough and hardy on the outside but soft on the inside in order to not cause discomfort to the worm itself. While explaining

the project to Fast Company, the Director of the Mediated Matter Group, Neri Oxman stated, "The silkworm embodies everything an additive fabrication system currently lacks. It's small in size and mobile in movement, it produces natural material of variable mechanical properties, and it spins a non-homogenous, non-woven textile-like structure."

While the silkworms were busy creating the dome, the MIT researchers studied their movements with the help of tiny magnets attached to the creatures' heads. The researchers hope to feed that information into a 3D printer so that in turn, it may be able to create non-homogenous structures just like the silkworms. In order to get the silkworms to create the dome, the researchers first built a computer model of the dome itself and then created a frame for the silkworms to work on with metal and thread. They also designed the frame to allow light through in deliberate patterns to influence the silkworms' movements.

The MIT researchers' project could be extremely fruitful in improving the overall efficiency and potential of 3D printers. Currently, most 3D printers can only move in three directions and create mostly homogenous items. However, with the help of the data generated by the silkworms, 3D printers could be programmed to behave more like the silkworms and break their current shackles.

The 'Silk Pavilion' project could itself be self-replicating since the silkworms involved are changing into moths and have the potential of laying 1.5 million eggs that could help in making 250 similar domes. Check out the very interesting video on the 'Silk Pavilion' project here: <http://dgit.in/18xMv14>

MIT Director uses a robot and a tablet to become omniscient, omnipresent

The Director of MIT's Media Lab, Joi Ito has exploited a new telepresence system installed by Double Robotics to be able to make his presence felt everywhere on the campus. The Double Robotics telepresence system employs a wheeled robot to which a tablet can be attached. The robot can then be controlled using an app or through the Web.



MIT Media Lab Director Joi Ito using the Double Robotics telepresence system

A video posted on the MIT Media Lab's Facebook page shows Joi Ito 'rolling' around the Lab's office building, albeit a little slowly, and interacting with the people milling around the robot. While the entire video shows that the technology is still in its nascent stages, it does show a lot of potential. We can easily see a telepresence system, aided by robotics to offer mobility, gaining credence among big companies whose executives can't always afford to travel to every single one of their branches.

You can check out Joi Ito's adventures with the Double Robotics' telepresence system in the video here: <http://dgit.in/14ijNh8>